

Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ of a single variable, concave on $[\ell_1, u_1]$, where $\ell_1 < u_1$. Consider also the function g of two variables, defined by $g(x_1, x_2) = f(x_1)$ on $A = [\ell_1, u_1] \times [\ell_2, u_2]$, where $\ell_2 < u_2$. Show that g is concave on A .

Remember that g is concave on a convex set A if

$$g((1 - \lambda)x + \lambda y) \geq (1 - \lambda)g(x) + \lambda g(y)$$

for all $x, y \in A$ and for all $\lambda \in [0, 1]$.